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Editor-in-Chief  
Expert Systems with Applications

Dear Editor,

Please consider the manuscript, “Reinforcement learning with forecast-loss rewards for sensor scheduling under power constraints,” for publication as a Research Article in *Expert Systems with Applications*.

The paper presents the Prediction-Driven Proximal Policy Optimization (PD-PPO) scheduler for a power budget that prevents all channels from operating simultaneously. Its complete training configuration combines a reward based on forecast loss, behavior cloning pretraining, continuing advantage-weighted behavior cloning, an auxiliary future-condition classification loss, and simulated warning signals. A feasibility layer checks the steady-state and startup-peak power limits, while the minimum dwell-time constraint determines the executed channel subset. All six candidates satisfy both power limits in the reported  $q = 1$  instance, so the optional-channel cardinality constraint and minimum dwell-time constraint are the active restrictions.

Across 22 post-selection evaluation seeds in a controlled six-channel cold-region monitoring simulation, the complete PD-PPO configuration achieved 10.6% lower mean forecast loss and 8.0% lower macro-averaged normalized forecast loss than the validation-selected fixed-schedule baseline. Both losses are lower in every seed and remain lower over each continuous 5,242-step scoreable test partition. The separate 24-seed descriptive aggregate, which includes two pilot/model-selection seeds, retains the 10.1% and 7.9% reductions. The evaluation also includes matched PPO training configurations with different scalar objectives, a Double DQN training-configuration comparison, an alternative-forecaster analysis, and executed-action diagnostics. The privileged one-step look-ahead reference uses next-step test targets. The privileged true-label baseline uses the exact label sequence  $c_t, \dots, c_{t+16}$  at each decision; neither information source is available to a deployed scheduler.

The manuscript fits the journal’s scope through constrained sensor scheduling, forecast-based evaluation, explicit information boundaries, and reproducible system-level analysis. It is original, is not under consideration elsewhere, and has been approved by both authors. The authors declare no competing interests.

The code and evidence repository will be made public upon acceptance. No graphical asset was created or modified using a generative image tool. The anonymous manuscript, title page, highlights, supplementary material, and reproducibility archive are separate submission files.

Sincerely,

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